

Summer Math & Scratch Coding Camp

Instructor: Mason Cao (Lambert High School)

Location: Sharon Forks Library - Conference Room

Schedule: Weekly for 6 Weeks (2.5 Hour Sessions)

Ages: 4th to 8th Grade

Email: masoncao7@gmail.com | Phone: 404-567-4668

Welcome!

Hi parents and students! Welcome to the Summer Math & Coding Camp. I designed this 6-week program to help students bridge their elementary and middle school math skills with foundational computer science. During our 2.5-hour sessions, we spend the first half diving deep into mathematical concepts (like coordinate graphs and number variables) and the second half applying those exact concepts to build complex, interactive games in MIT's Scratch platform.

What You Need to Bring

- A laptop (Windows, Mac, or Chromebook) with a charger.
- A modern web browser (like Google Chrome).
- A free account set up on **scratch.mit.edu**.
- *Prerequisites:* Students should be comfortable with basic arithmetic and eager to learn logic-based problem solving!

Classroom Expectations

- **Collaboration:** We will have dedicated "Debugging Sessions" where students present broken code to the class and we collaboratively figure out how to fix it.
- **Food and Drinks:** To protect the library's space, please keep all snacks and water bottles on the side tables, away from the keyboards.
- **Attendance:** Because our 2.5-hour sessions cover a significant amount of material that builds cumulatively, please try your best not to miss a week!

Weekly Schedule

Week	Math & Logic Focus	Scratch Application
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Week 1	Coordinate Math & Screen Movement Mastering the 4-quadrant math graph (X and Y axes), plotting points, and calculating distance so characters move exactly where we want.	Platformer Game (Part 1): We will code our own jumping game, figuring out the math to make a character jump up and fall back down smoothly without falling through the floor.
Week 2	Algebraic Variables & Tracking Using placeholders for numbers that change, tracking data, and understanding inequalities ($>$ $<$ $=$).	Tycoon Simulator: Students will build an economy game where they track multiple resources (money, health, time) at the same time, making sure the numbers update correctly when buying items.
Week 3	Logic & Conditionals Evaluating True/False statements, compound rules (AND/OR), and nested "If/Then" structures.	Smart Enemy Movement: Using nested "If/Then/Else" blocks, students will program an enemy character that can "see" the player, move around simple walls, and make choices based on how far away the player is.
Week 4	Geometry & Loops Calculating interior/exterior angles of shapes, understanding perimeter, and using endless loops.	Auto-Drawing Levels: Using "Repeat" loops and the Pen tool, students will write code that automatically draws complicated, randomized maze layouts or cool geometric patterns.
Week 5	Step-by-Step Logic & Timing Understanding order of operations, having multiple things happen at once, and triggering events.	Multi-Level Games: We will master "Broadcast" messages to switch smoothly between game screens (Start Menu -> Level 1 -> Game Over), keeping track of different background tasks at the same time.
Week 6	Probability & Final Assessment Random number ranges, calculating chances, and step-by-step problem solving.	Capstone Project: A fully independent 2.5-hour build session. Students must design a game incorporating coordinate movement, at least three tracked variables, and complex "If/Then" logic. The session ends with a parent showcase!